Continuity

$f(x)$ is **continuous** at $x = a$:

① $\lim_{x \to a} f(x)$ exists $\Rightarrow \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

② $f(x)$ is defined $\Rightarrow a \in D_f$.

③ $\lim_{x \to a} f(x) = f(a)$

} Continuity Test.

Example 1

Given $f(x) = \begin{cases} x+1, & x \le 3 \\ 2x-2, & x > 3 \end{cases}$, is the function f(x) continuous at $x = 3$?

piecewise defined function

Continuous?
$x = ??$
reason?

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (x+1) = 3+1 = 4$$

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (2x-2) = 2(3)-2 = 4$$

$\therefore \lim_{x \to 3} f(x) = 4$

$$\lim_{x \to 3} f(x) = f(3) = 4 \quad \checkmark$$

$\therefore f(x)$ is continuous at $x = 4$ ✌

at $x = 3$; $f(x) = x+1$

$f(3) = 4$

Example 2

At what value/values of x is the function $h(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$ continuous?

$h(x)$ is undefined when $x^2 - 5x + 6 = 0$

$(x-2)(x-3) = 0$

$x = 2, x = 3$

At all $x = a$ other than $x = 2, x = 3$: $\lim_{x \to a} \frac{(x-3)(x+3)}{(x-2)(x-3)} = \frac{a+3}{a-2}$ ✓

$\therefore f(x)$ is not continuous at $x = 2$ and $x = 3$

Example 3

The functions $f(x) = x^4 + 20$ and $g(x) = 5x(x - 2)$ are continuous at every value of x. Find the values for which the function $\frac{f(x)}{g(x)}$ is continuous.

Q

→ rational function

polynomial

degree = 4

degree = 2

$$\frac{f(x)}{g(x)} = \frac{x^4+20}{5x(x-2)}$$

Let $x = a$;

$$\lim_{x \to a} f(x) = a^4 + 20$$

$$= f(a)$$

$$\lim_{x \to a} g(x) = 5a(a-2)$$

$$= g(a)$$

$\frac{f(x)}{g(x)}$ is undefined at $x = 0$ and $x = 2$.

$\therefore \frac{f(x)}{g(x)}$ is discontinuous at $x = 0$ and $x = 2$

Example 4

Find the discontinuity point(s) for the following functions.

(a) $f(x)=\dfrac{3x+1}{x^2+7x-2}$

(b) $h(x)=\dfrac{x^2-9}{x^2-5x+6}$

(c) $g(x)=\dfrac{x^2+3x}{x+3}$

} rational functions ↓

(a) $x^2+7x-2=0$

$$x=\frac{-7\pm\sqrt{7^2-4(1)(-2)}}{2(1)}$$

$$=\frac{-7\pm\sqrt{57}}{2}$$ #

(b) $x^2-5x+6=0$

$(x-2)(x-3)=0$

$x=2,3$ #

(c) $x+3=0$

$x=-3$ #

Example 5

What is the value for constant k that will make the function $f(x)=\begin{cases} kx^2 & , \ x \le 2 \\ 2x+k & , \ x > 2 \end{cases}$ continuous on $(-\infty,+\infty)$?

$x \in \mathbb{R}$

$x = 2$??

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} kx^2$$

$$= 4k$$

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (2x+k)$$

$$= 4+k$$

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$$

$$4k = 4+k$$

$$k = \frac{4}{3}$$

Example 6

For which value/s of x is $f(x) = \begin{cases} 2-3x, & x \le -1 \\ x^2 - x + 3, & x > -1 \end{cases}$ is not continuous?

$x = -1$?

$\lim_{x \to -1^-} (2-3x) = 2 - 3(-1) = 5$

$\lim_{x \to -1^+} (x^2 - x + 3) = (-1)^2 - (-1) + 3 = 5$

$\lim_{x \to -1} f(x) = 5$

at $x = -1$; $f(x) = 2-3x$

$f(-1) = 2-3(-1)$

$= 5$

$\therefore$ no value of x because f(x) is always continuous

$\therefore f(x)$ is continuous at $x = -1$.